

AI-based Generalizable Model (GzMPDD) to Detect Plant Leaf Disease Using YOLOv8n

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Abstract

Plant diseases remain a major challenge to global agriculture, often resulting in significant losses for farmers and threatening food security. Many existing AI tools struggle to detect diseases across diverse plant types due to poor generalization and sensitivity to environmental noise. This work presents a generalizable, real-time disease detection system using YOLOv8, designed to help both farmers and professionals. By combining robust background removal and a dual-dataset approach, the model accurately identifies diseases from leaf images across various species. Tested on 19 classes covering 7 plants and 12 diseases, the system demonstrates improved adaptability and reliability compared to traditional and earlier AI-based methods[9, 6, 11, 14].

Keywords: Generalizable Model (GzMPDD), Convolutional Neural Network (CNN), Annotation, YOLOv8, Segmentation, Roboflow, Ultralytics

1. Introduction

Efficient farming and healthy crops are vital for sustainable agriculture and meeting global food demands. Undetected plant diseases can cause significant yield loss and economic damage, threatening food security. Therefore, timely and accurate disease identification alongside effective management is essential. While traditional manual inspections are often slow and error-prone, modern automated techniques based on deep learning, including

Convolutional Neural Networks (CNNs), have shown promise[9, 6, 13, 11]. However, these methods face challenges such as poor generalization across diverse plant species and sensitivity to environmental factors[15, 5].

This paper presents a two-step approach using the advanced YOLOv8 model to improve plant disease detection. It first removes background noise with pre-trained segmentation to focus on disease features[5]. Then, it uses a dual dataset strategy for reliable detection across different plant species. The lightweight and fast YOLOv8 model enables real-time, mobile-friendly detection, ideal for practical use in the field[16, 15].

The key contributions of this work are as follows:

- The introduction of a segmentation-based pre-processing step to improve disease feature extraction[12, 5].
- A dual-dataset training approach to enhance model generalization across species[1, 2].
- The use of YOLOv8 for real-time, mobile-compatible disease detection, ensuring robustness in field conditions[15, 16], and
- Separately labeling the plant and the disease, this methodology trains the model to detect similar symptoms in taxonomically related species, making it more adaptable to real-world agricultural challenges[10].

This innovative combination of dataset structure, segmentation, and YOLOv8 integration creates a practical and scalable solution for modern agriculture, enabling early and efficient disease management while promoting sustainable practices [11, 2, 14].

2. Related Work

The identification of plant diseases is a key research area in agriculture and computer vision. Detection methods are broadly classified into non-learning-based techniques, which rely on hand-crafted feature extraction, and learning-based methods, which utilize machine learning for improved accuracy[9, 13, 3].

2.1. Traditional Feature-Based Methods for Plant Disease Detection

Traditional methods for detecting plant disease involve hand-crafted feature extraction techniques. While these methods are efficient in terms of computation and somewhat intuitive, they have limitations, particularly when applied to large datasets due to their complexity and variability. One such

technique is Local Binary Patterns (LBP) and Histogram of Oriented Gradients (HOG), commonly used to extract textural and structural features. However, these methods struggle with larger or more varied datasets, leading to reduced accuracy [2, 7, 13]. Scale-Invariant Feature Transform (SIFT) and Speeded-Up Robust Features (SURF) focus on significant visual features but are sensitive to scale, rotation, and lighting [9, 6, 8].

2.2. Machine Learning-Based Methods for Plant Disease Detection

Advancements in machine learning have improved the accuracy and scalability of plant disease detection by enabling models to learn complex patterns from image datasets. Convolutional Neural Networks (CNNs) are widely used for leaf disease identification due to their strong image-analysis and pattern-recognition capabilities [14, 6, 4]. They can extract detailed disease features from large sets of leaf images [3, 8], and have been applied successfully in tasks such as grapevine disease detection [2]. However, CNNs require large, well-annotated datasets, limiting their adaptability to new crops or unseen conditions [11].

YOLO-based models are another major development, combining high accuracy with fast real-time processing. YOLOv5 has been used to detect diseases across multiple crop types, generating rapid and actionable insights for farmers. The latest version, YOLOv8, offers further improvements in accuracy and speed, and its lightweight design allows deployment on mobile devices, making advanced detection accessible even in remote agricultural areas [9, 2].

3. Methodology

The proposed system takes a picture of a plant and its species name as the input, diagnoses the disease, pinpoints affected areas, and reports the outcome, streamlining disease diagnosis for agricultural experts. The model is trained on two datasets—a primary set covering ten species with two diseases each and an auxiliary set with related diseases across species. Pre-processing includes data augmentation: rotations, flips, brightness changes, noise injection, and cropping. Evaluation yields bounding boxes and confidence scores in disease detection.

3.1. Framework Overview

The system is structured into three phases:

- **Background Removal:** Use segmentation to isolate leaves from the background, reducing noise and improving detection accuracy.
- **Disease Detection:** YOLOv8 analyzes processed images for disease symptoms.
- **Cross-Species Disease Mapping:** Auxiliary dataset allows mapping and recognition of symptoms in species not present in the primary data [1].

3.2. Algorithm

1. Input Acquisition: Accept image and plant name.
2. Background Removal: Pre-process using segmentation models.
3. Image Recognition: Compare against known diseases.
4. Decision Point: If match, save; else, run auxiliary algorithm.
5. Auxiliary Matching: Attempt to identify based on auxiliary dataset.
6. Output: Return plant and disease identified, or "No Match Found".

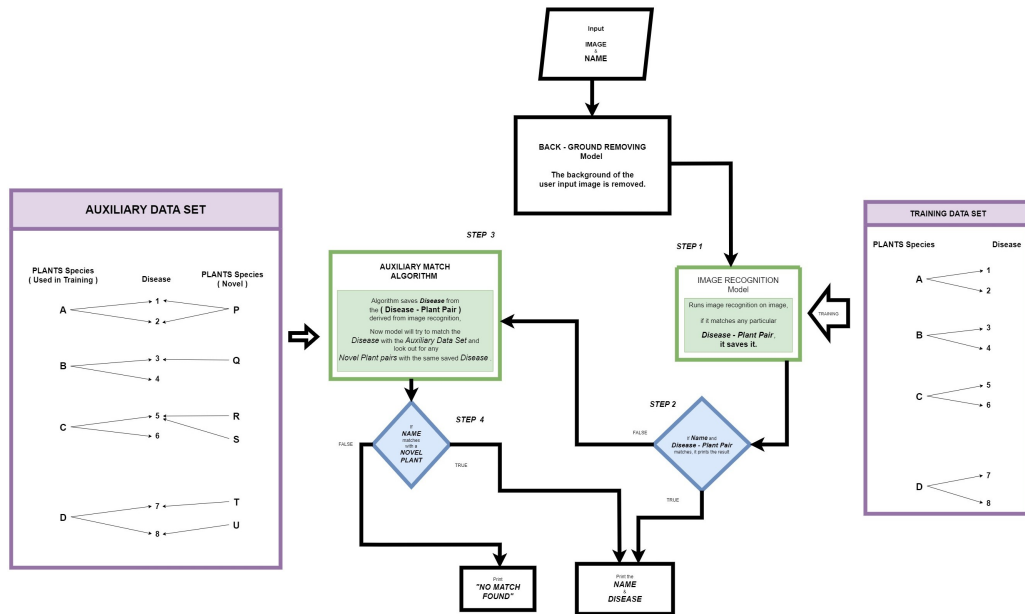


Figure 1: Algorithm Architecture.

3.3. Dataset Description

Primary Dataset: Using custom annotated dataset of 4,221 images (7 species, 12 diseases), resolution 640×640 pixels. Data annotation via Roboflow, exported in YOLO format. Augmentation: rotations (90°), flips, brightness (-10% to $+10\%$), noise (0.53%), cropping. Training/validation/testing split: 91%/6%/2%. See Table 1 for labels:

Table 1: Dataset Plant-Disease Mapping

Plants	Diseases
Apple	Scab, Black Rot, Rust
Squash	Powdery Mildew
Tomato	Bacterial Spot, Spider Mite Leaf Spot, Mosaic, Yellow Leaf Curl, Late Blight
Potato	Late Blight
Pepper	Bacterial Spot
Orange	Huanglongbing
Grapevine	Esca, Black Rot Gb

Auxiliary Dataset: Additional species with similar disease symptoms, prepared for generalization.

3.4. Training Setup

Trained using NVIDIA RTX 3070 GPU (12GB VRAM), Batch Size 16, Epochs: 246.

4. Result

4.1. Experiment Results

Table 2: Model Performance at Epochs

Epoch	Precision (%)	Recall (%)	F1 Score (%)
30	66.78	61.63	64.10
50	55.26	68.74	61.27
125	51.65	65.50	57.75
246	54.55	56.83	55.66

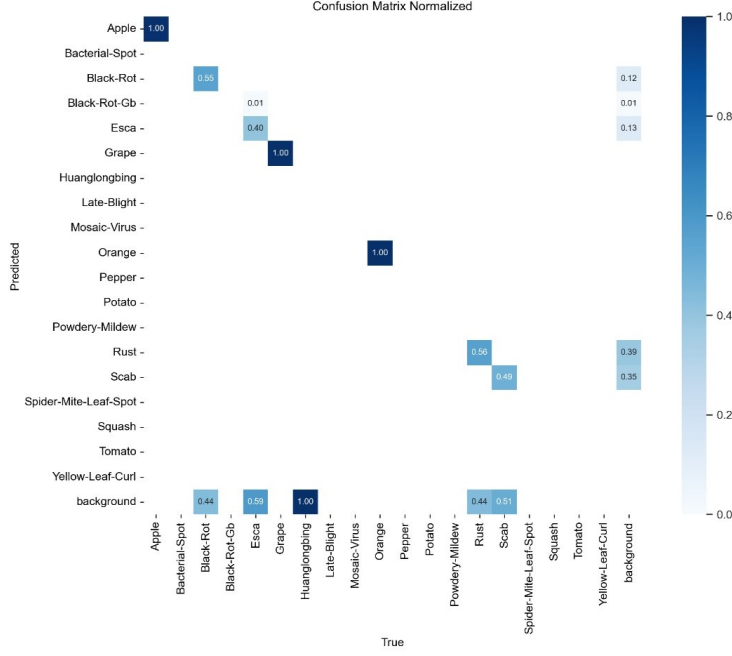


Figure 2: Normalized Confusion Matrix

4.2. Results and Comparison

Table 3: Comparison With Existing Work

Model Used	Precision	Recall	F1 Score	mAP50	mAP50-95
YOLOv8-ACCW	—	—	92.4	92.8	73.8
Existing [9]	35.22	33.78	34.48	—	—
EfficientNetB0	98.27	98.26	98.26	—	—
GzMPDD (this paper)	54.54	56.83	55.65	55.52	44.41

5. Conclusion

This study introduces a novel and generalizable AI-based plant disease detection system leveraging YOLOv8. The system demonstrates enhanced performance through innovative architecture that integrates background removal, dual-dataset methodology, and segmentation techniques.

Though the experiment results are lower as compared to the existing approaches in models like EfficientNet, this approach has great potential. The

work opens door for exploring new methods for Disease Detection in Leaves. The model’s accuracy can be further enhanced by training on a larger dataset. Moreover the method provides a platform to integrate Reinforcement Learning through a feedback loop for model to train on new Plant Species-Disease pairs, thus enabling the model to expand its horizon to more plants and their diseases.

References

- [1] Vishal Singh Bhati, Namita Tiwari, and Meenu Chawla. A generalized zero-shot deep learning classifier for emotion recognition using facial expression images. *IEEE Access*, 2025.
- [2] Zuxing Chen, Junjie Feng, Kun Zhu, Zhenyan Yang, Yanhong Wang, and Mingyue Ren. Yolov8-accw: Lightweight grape leaf disease detection method based on improved yolov8. *IEEE Access*, 12:123595–123608, 2024.
- [3] Jennifer Eunice, Daniela Elena Popescu, M Kalpana Chowdary, and Jude Hemanth. Deep learning-based leaf disease detection in crops using images for agricultural applications. *Agronomy*, 12(10):2395, 2022.
- [4] Konstantinos P Ferentinos. Deep learning models for plant disease detection and diagnosis. *Computers and electronics in agriculture*, 145:311–318, 2018.
- [5] P Goncharov, G Ososkov, A Nechaevskiy, A Uzhinskiy, and I Nestsiairenia. Disease detection on the plant leaves by deep learning. In *International Conference on Neuroinformatics*, pages 151–159. Springer, 2018.
- [6] Aissam El Ibrahimi and Nabil El Akchioui. A review on plant diseases detection using artificial intelligence techniques. In *INTERNATIONAL CONFERENCE ON ADVANCES IN COMMUNICATION TECHNOLOGY AND COMPUTER ENGINEERING*, volume 2814, page 040018. AIP Publishing LLC, 2023.
- [7] Liangquan Jia, Tao Wang, Yi Chen, Ying Zang, Xiangge Li, Haojie Shi, and Lu Gao. Mobilenet-ca-yolo: An improved yolov7 based on the mobilenetv3 and attention mechanism for rice pests and diseases detection. *Agriculture*, 13(7):1285, 2023.

- [8] Zong-wei Jia, Jing Hao, Yi-ming Hou, Rui-bin Wang, Ru-yi Zhang, Simin Yao, Ju Zhang, Hao Ke, and Yi Shao. Study on rapid detection and identification of multi category apple leaf disease. 2022.
- [9] Emmanuel Moupojou, Florent Retraint, Hyppolite Tapamo, Marcellin Nkenlifack, Cheikh Kacpah, and Appolinaire Tagne. Segment anything model & fully convolutional data description for plant multi-disease detection on field images. *IEEE Access*, 2024.
- [10] Mamillapally Nagaraju and Priyanka Chawla. Systematic review of deep learning techniques in plant disease detection. *International journal of system assurance engineering and management*, 11(3):547–560, 2020.
- [11] Natasha Nigar, Hafiz Muhammad Faisal, Muhammad Umer, Olukayode Oki, and Jose Manappattukunnel Lukose. Improving plant disease classification with deep-learning-based prediction model using explainable artificial intelligence. *IEEE Access*, 12:100005–100014, 2024.
- [12] Nengyuan Pan, Weiming Yang, Yuting Luo, and Yonglin Wang. Identification of leaf disease based on memristor convolutional neural networks. *IEEE Access*, 2024.
- [13] Zafar Salman, Abdullah Muhammad, Md Jalil Piran, and Dongil Han. Crop-saving with ai: latest trends in deep learning techniques for plant pathology. *Frontiers in Plant Science*, 14:1224709, 2023.
- [14] Muhammad Shoaib, Babar Shah, Shaker Ei-Sappagh, Akhtar Ali, Asad Ullah, Fayadh Alenezi, Tsanko Gechev, Tariq Hussain, and Farman Ali. An advanced deep learning models-based plant disease detection: A review of recent research. *Frontiers in Plant Science*, 14:1158933, 2023.
- [15] Abhishek Thakur, Sangeeth Venu, and Muralimohan Gurusamy. An extensive review on agricultural robots with a focus on their perception systems. *Computers and Electronics in Agriculture*, 212:108146, 2023.
- [16] Xiang Yin, Wenhua Li, Zhen Li, and Lili Yi. Recognition of grape leaf diseases using mobilenetv3 and deep transfer learning. *International Journal of Agricultural and Biological Engineering*, 15(3):184–194, 2022.